Fall 1980

Problem 1 (Fa80). Define $F(x) = \int_{\sin x}^{\cos x} e^{(t^2+xt)}\,\mathrm{d}t$. Compute $F'(0)$.

Problem 2 (Fa80, Fa92). Are the matrices give below similar ?

$$A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 1 \\ -1 & 0 & 2 \end{pmatrix} \qquad \text{and} \qquad B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Problem 3 (Fa80). Do there exist functions $f(z)$ and $g(z)$ that are analytic at $z = 0$ and that satisfy

1. $f\left(\frac{1}{n}\right) = f\left(-\frac{1}{n}\right) = \frac{1}{n^2}, n = 1, 2, \ldots$
2. $g\left(\frac{1}{n}\right) = g\left(-\frac{1}{n}\right) = \frac{1}{n^3}, n = 1, 2, \ldots$?

Problem 4 (Fa80). Let G be the group of orthogonal transformations of $\mathbb{R}^3$ to $\mathbb{R}^3$ with determinant 1. Let $v \in \mathbb{R}^3$, $\|v\| = 1$, and let $H_v = \{T \in G \mid Tv = v\}$.

1. Show that H_v is a subgroup of G.
2. Let $S_v = \{T \in G \mid T$ is a rotation of $180°$ about a line orthogonal to $v\}$. Show that S_v is a coset of H_v in G.

Problem 5 (Fa79, Fa80, Sp85, Su85, Fa96, Fa98, Sp99, Sp26). Prove that

$$\int_0^\infty \frac{x^{\alpha-1}}{1+x}\,\mathrm{d}x = \frac{\pi}{\sin \pi\alpha}$$

What restrictions must be placed on α?

Problem 6 (Fa80). Let $M_{2\times 2}$ be the ring of real 2×2 matrices and $S \subset M_{2\times 2}$ the subring of matrices of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

1. Exhibit an isomorphism between S and $\mathbb{C}$.

2. Prove that

$$A = \begin{pmatrix} 0 & 3 \\ -4 & 1 \end{pmatrix}$$

lies in a subring isomorphic to S.

3. Prove that there is an $X \in M_{2\times 2}$ such that $X^4 + 13X = A$.

Problem 7 (Fa80, Sp13). Let g be 2π-periodic, continuous on $[-\pi, \pi]$ and have Fourier series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Let f be 2π-periodic and satisfy the differential equation

$$f''(x) + kf(x) = g(x)$$

where $k \neq n^2, n = 1, 2, 3, \ldots$. Find the Fourier series of f and prove that it converges everywhere.

Problem 8 (Fa80). Let $\mathbb{F}_2 = \{0, 1\}$ be the field with two elements. Show that $\mathrm{GL}_2(\mathbb{F}_2)$, the group of invertible 2×2 matrices with entries in $\mathbb{F}_2$ is isomorphic to S_3.

Problem 9 (Fa80). For a real 2×2 matrix

$$X = \begin{pmatrix} x & y \\ z & t \end{pmatrix}$$

let $\|X\| = x^2 + y^2 + z^2 + t^2$, and define a metric by $d(X, Y) = \|X - Y\|$. Let $\Sigma = \{X \mid \det X = 0\}$. Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

Find the minimum distance from A to Σ and exhibit an $S \in \Sigma$ that achieves this minimum.

Problem 10 (Fa80). Show that there is an $\varepsilon > 0$ such that if A is any real 2×2 matrix satisfying $|a_{ij}| \leqslant \varepsilon$ for all entries a_{ij} of A, then there is a real 2×2 matrix X such that $X^2 + X^{\mathsf{T}} = A$. Is X unique?

Problem 11 (Fa80). Let $f(z)$ be an analytic function defined for $|z| \leqslant 1$ and let

$$u(x, y) = \Re f(z) \qquad z = x + iy$$

Prove that

$$\int_C \frac{\partial u}{\partial y}\,\mathrm{d}x - \frac{\partial u}{\partial x}\,\mathrm{d}y = 0$$

where C is the unit circle, $x^2 + y^2 = 1$.

Problem 12 (Fa80). Prove that any group of order 6 is isomorphic to either $\mathbb{Z}_6$ or S_3 (the group of permutations of three objects).

Problem 13 (Fa80). Let $f(x) = 1/4 + x - x^2$. For any real number x, define a sequence (x_n) by $x_0 = x$ and $x_{n+1} = f(x_n)$. If the sequence converges, let x_∞ denote the limit.

1. For $x = 0$, show that the sequence is bounded and nondecreasing and find $x_\infty = \lambda$.

2. Find all $y \in \mathbb{R}$ such that $y_\infty = \lambda$.

Problem 14 (Fa80). Exhibit a set of 2×2 real matrices with the following property: A matrix A is similar to exactly one matrix in S provided A is a 2×2 invertible matrix of integers with all the zeros of its characteristic polynomial on the unit circle.

Problem 15 (Fa80). Consider the differential equation $x'' + x' + x^3 = 0$ and the function $f(x, x') = (x + x')^2 + (x')^2 + x^4$.

1. Show that f decreases along trajectories of the differential equation.

2. Show that if $x(t)$ is any solution, then $(x(t), x'(t))$ tends to $(0, 0)$ as $t \to \infty$.

Problem 16 (Fa80, Sp96). Suppose that A and B are real matrices such that $A^\top = A$,

$$v^\top A v \geqslant 0$$

for all $v \in \mathbb{R}^n$ and

$$AB + BA = 0$$

Show that $AB = BA = 0$ and give an example where neither A nor B is zero.

Problem 17 (Fa79, Fa80). Let $\{P_n\}$ be a sequence of real polynomials of degree $\leqslant D$, a fixed integer. Suppose that $P_n(x) \to 0$ pointwise for $0 \leqslant x \leqslant 1$. Prove that $P_n \to 0$ uniformly on $[0, 1]$.

Problem 18 (Fa80). Suppose that f is analytic inside and on the unit circle $|z| = 1$ and satisfies $|f(z)| < 1$ for $|z| = 1$. Show that the equation $f(z) = z^3$ has exactly three solutions (counting multiplicities) inside the unit circle.

Problem 19 (Fa80, Sp10). Let X be a compact metric space and $f : X \to X$ an isometry. Show that $f(X) = X$.

Problem 20 (Fa80). Let R be a ring with identity. Call $x \in R$ a *unit* if $xy = yx = 1$ for some $y \in R$. Let R^* denote the set of units.

1. Prove R^* is a multiplicative group.
2. Prove that $\mathbb{Z}^*[i]$ is isomorphic to $\mathbb{Z}_4$.